College Football Bowl Predictor

UE24CS352A Machine Learning Mini-Project · Problem 88 · Team G-29: Likith Kumar K M (PES1UG24AM373), Uthkarsha (PES1UG24AM413) · Section G

1. Problem statement

Predict the winner of each college football bowl game and attach a **confidence** to each pick, following Stanford CS229 (2015) project 101. Confidence matters because bowl pick'em contests rank your picks: a correct pick earns its confidence rank as points, so a good model must know *which* games it is sure about.

2. Dataset

The original blog-sourced data is not available, so we rebuilt it from public play-by-play data (*sportsdataverse/cfbfastR-data*, GitHub): 16 seasons (2005-2020), ~12.6k games, ~25k team-games, aggregated from ~2M plays. We keep **566 bowl/postseason games**, of which 545 have two teams with ≥8 regular-season games. Per team-season we compute 28 features from **regular-season games only** (no leakage): scoring (PPG, PAPG), offensive and defensive yards/carry and yards/attempt, completion, sack and interception rates, EPA/play, success rate, havoc rate, turnover margin, penalties, punt average, FG%, and an end-of-season margin-aware **Elo rating** (K=20, home edge 55, 1/3 regression to the mean each year). A game is the difference of the two teams' features; the label is the score margin.

3. Approach

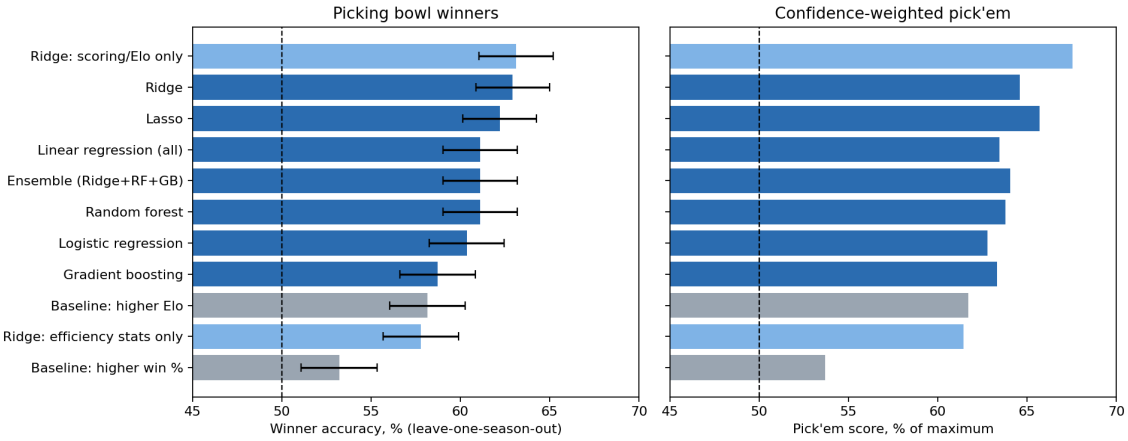
- **Symmetric formulation:** x = f(team1) – f(team2). Linear models have no intercept and are trained on randomly oriented pairs, so swapping teams exactly flips the prediction; tree models train on both orientations and are antisymmetrised.
- **Models:** OLS, Ridge, Lasso, logistic regression, random forest, gradient boosting, and an Ensemble, versus baselines (higher win %, higher Elo) and ablations (scoring/Elo features only; efficiency stats only).
- **Validation:** leave-one-season-out CV (16 folds) with regularisation tuned by nested season-wise CV inside each training set. Metrics: winner accuracy, pick'em score (% of maximum N(N+1)/2 per season), margin MAE.
- **Confidence:** |predicted margin|, mapped to a win probability by a one-parameter logistic calibration fit on out-of-fold predictions.

4. Implementation

Python (pandas, scikit-learn, matplotlib, Streamlit). `build_dataset.py` → `features.py` → `models.py/train.py` → `predictor.py` with a CLI (`predict.py`) and a Streamlit demo (matchup tab with per-feature explanation; season pick'em sheet). Full pipeline runs in ~3 minutes; the README has setup steps.

5. Results (545 bowl games, leave-one-season-out)

Model	Winner acc.	Pick'em (% of max)	Margin MAE
Coin flip (reference)	50.0% ± 2.1	50.0%	13.8
Baseline: higher win %	53.2% ± 2.1	53.7%	-
Baseline: higher Elo	58.2% ± 2.1	61.7%	-
Linear regression (all)	61.1% ± 2.1	63.4%	13.5
Ridge	62.9% ± 2.1	64.6%	13.2
Lasso	62.2% ± 2.1	65.7%	13.1
Logistic regression	60.4% ± 2.1	62.8%	-
Random forest	61.1% ± 2.1	63.8%	13.2
Gradient boosting	58.7% ± 2.1	63.3%	13.3
Ensemble (Ridge+RF+GB)	61.1% ± 2.1	64.0%	13.2
Ridge: scoring/Elo only	63.1% ± 2.1	67.5%	12.9
Ridge: efficiency stats only	57.8% ± 2.1	61.5%	13.4



6. Conclusions

- **Ridge reaches 62.9% winner accuracy** (SE ≈ 2.1 pts) against 58.2% for the Elo baseline and 53.2% for win %. The gain over Elo is statistically suggestive (exact sign test, p = 0.029) but is one of five comparisons, so it is not conclusive.
- **Simple beats flexible.** With only ~550 games and 28 correlated features, regularised linear models match or beat random forest (61.1%), boosting (58.7%) and the ensemble (61.1%). Plain OLS (61.1%) is below Ridge, which shows shrinkage helps.
- **Scoring/Elo features carry the signal.** Ridge on only PPG, PAPG, margin, win % and Elo scores 63.1% (67.6% pick'em); advanced efficiency stats alone score 57.8%. The extra play-by-play features did not add measurable accuracy.
- **Confidence is only partly reliable.** Accuracy by confidence quartile is 64%, 59%, 56%, 73% (lowest to highest): the top quartile is clearly best, but the curve is not monotone, so the win-probability numbers should be read as rough.
- **Context.** The CS229 report cites ~66% on 2000-2010 data with different features; our data, seasons and protocol differ, so the numbers are not directly comparable. Bowl games have opt-outs, transfers and motivation effects that season statistics cannot see (*future work*: roster availability, betting lines, upset detection).

Code: private GitHub repository (shared with faculty/TAs). Data: sportsdataverse/cfbfastR-data (public).